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# Atrioventricular Nodal Reentry Tachycardia

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## Continuing Education Activity

Atrioventricular nodal reentrant tachycardia (AVNRT) represents the most common form of paroxysmal supraventricular tachycardia in adults and frequently presents with symptomatic palpitations in emergency and outpatient settings. The arrhythmia arises from dual atrioventricular nodal pathways that facilitate reentry, producing a regular narrow-complex tachycardia. Clinical manifestations include palpitations, lightheadedness, dyspnea, and, less commonly, hemodynamic instability. Although generally benign, delayed recognition or misclassification may result in recurrent symptoms, increased health care utilization, and missed opportunities for definitive therapy. Accurate electrocardiographic interpretation and timely management remain essential to optimize outcomes and reduce complications.

This educational activity reviews the mechanisms, clinical features, electrocardiographic findings, diagnostic evaluation, and differential diagnosis of AVNRT, as well as evidence-based acute and long-term management strategies. Participants learn to improve diagnostic accuracy, apply appropriate pharmacologic and nonpharmacologic interventions, and recognize indications for catheter ablation. Emphasis is placed on identifying complications and risk factors while enhancing patient-centered care. Collaboration with an interprofessional team strengthens care coordination, supports timely intervention, and improves symptom control, safety, and overall clinical outcomes.

**Objectives:**

- Identify the clinical and electrocardiographic features that support the diagnosis of atrioventricular nodal reentrant tachycardia.
- Apply evidence-based acute treatment strategies for atrioventricular nodal reentrant tachycardia, including vagal maneuvers, pharmacologic therapy, and synchronized cardioversion when indicated.
- Interpret the risks, benefits, and potential complications of medical therapy and catheter ablation when developing patient-centered treatment plans.
- Collaborate with nursing staff, pharmacists, and electrophysiology specialists to improve diagnostic accuracy, treatment selection, and coordinated care for patients with atrioventricular nodal reentrant tachycardia.

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## Introduction

The atrioventricular node is a subendocardial structure located in the inferoposterior right atrium. In an average adult, the atrioventricular node measures approximately 1 × 3 × 5 mm and sits within an anatomic region bordered posteriorly by the coronary sinus ostium, superiorly by the tendon of Todaro, and anteriorly by the septal tricuspid

valve annulus. This anatomic region is also commonly referred to as the triangle of Koch. The blood supply to the atrioventricular node arises from the atrioventricular nodal branch of the right coronary artery (90%) or the left circumflex artery (10%), depending on whether the heart has right- or left-coronary dominance. The first septal perforator of the left anterior descending artery also supplies blood to the atrioventricular node.[1][2]

Atrioventricular nodal reentrant tachycardia (AVNRT) is the most common form of paroxysmal supraventricular tachycardia in adults and a frequent cause of symptomatic palpitations presenting to emergency departments and outpatient cardiology clinics. The arrhythmia is characterized by a reentrant circuit that involves functionally distinct pathways within or adjacent to the atrioventricular node, resulting in a rapid, regular tachycardia with narrow QRS complexes in most cases. AVNRT is typically abrupt in onset and termination and may occur in otherwise healthy individuals or in those with underlying structural heart disease.[3]

Clinically, AVNRT is significant because it can cause symptoms such as palpitations, dizziness, chest discomfort, dyspnea, and presyncope or syncope. Although the arrhythmia is usually not life-threatening, recurrent episodes can impair quality of life and occasionally lead to hemodynamic compromise, particularly in older adults or those with limited cardiac reserve. Advances in electrophysiology have clarified the underlying mechanisms of AVNRT and enabled highly effective curative therapy with catheter ablation.[4][5] Electrocardiographic findings based on the reentrant circuits are as follows:

**Typical AVNRT (Slow-Fast AVNRT)**

Typical AVNRT has a very short RP interval, and P waves appear as:

- A pseudo-S wave in leads II, III, and aVF (see **Image**. Electrocardiogram of Typical AVNRT With Pseudo-S Waves)
- A pseudo-R' wave in lead V1

**Atypical AVNRT (Fast-Slow or Slow-Slow AVNRT)**

- Variable RP interval with discrete P waves following the QRS complex (see **Image**. Electrocardiogram of a Long RP Narrow Complex Tachycardia)
- P waves between the QRS complex and T waves (QRS-P-T complexes), or late P waves after a QRS complex, often mimicking atrial tachycardia

## Etiology

The etiology of AVNRT is primarily related to the presence of dual conduction pathways within or adjacent to the atrioventricular node. These pathways typically consist of a fast pathway, characterized by rapid conduction but a longer refractory period, and a slow pathway, which conducts more slowly but has a shorter refractory period.[6] This dual physiology allows the formation of a reentrant circuit when a premature atrial impulse encounters a refractory fast pathway and conducts through the slow pathway. Once the fast pathway recovers excitability, the impulse can travel retrogradely through it, completing a reentrant loop and initiating tachycardia.

Although the exact cause of dual atrioventricular nodal physiology is not fully understood, evidence suggests that it is related to congenital variations in the architecture of atrioventricular nodal tissue and to transitional cells surrounding the atrioventricular node. Structural heart disease is uncommon in most patients with AVNRT; however, conditions that increase atrial ectopy or alter autonomic tone, such as stress, stimulants, or underlying cardiac disease, may facilitate the initiation of the arrhythmia. Genetic predisposition and age-related changes in conduction tissue have also been suggested to contribute.[7]

## Epidemiology

AVNRT is the most common form of paroxysmal supraventricular tachycardia, accounting for approximately 50% to 60% of all supraventricular tachycardia cases encountered in clinical practice.[8] The overall prevalence of supraventricular tachycardia in the general population is estimated at 2.25 per 1000 individuals, with an annual incidence of approximately 35 cases per 100,000 persons. AVNRT can occur at any age, but it most commonly presents in young adults, although some individuals may remain asymptomatic until later decades of life.[9] Female predominance is consistently observed, with roughly two-thirds of cases occurring among women.[10]

In contrast to some other arrhythmias, most patients with AVNRT have structurally normal hearts, and the condition is often seen in otherwise healthy individuals. Familial clustering has also been described, suggesting a possible genetic predisposition, with first-degree relatives having a higher likelihood of developing AVNRT than the general population. Despite its frequent occurrence, AVNRT generally carries a benign prognosis, particularly when appropriate therapy such as catheter ablation is available. Understanding its epidemiologic patterns is important for clinicians because AVNRT represents a substantial proportion of symptomatic tachyarrhythmias evaluated in emergency departments and electrophysiology practices worldwide.

## Pathophysiology

These pathways typically consist of a fast pathway, which conducts impulses rapidly but has a relatively long refractory period, and a slow pathway, which conducts more slowly but has a shorter refractory period. Under normal sinus rhythm, atrial impulses preferentially conduct through the fast pathway to the His–Purkinje system. However, when a premature atrial contraction occurs, the fast pathway may still be refractory, forcing the impulse to travel antegrade through the slow pathway.

Because conduction through the slow pathway is delayed, the fast pathway has time to recover excitability. The impulse can then propagate retrogradely up the fast pathway back to the atria, creating a self-sustaining reentrant circuit. This continuous loop of antegrade conduction through the slow pathway and retrograde conduction through the fast pathway results in a rapid, regular tachycardia, typically with rates between 140 and 250 beats per minute. This form, known as typical (slow-fast) AVNRT, accounts for the majority of cases.[2][11]

Less common variants include atypical AVNRT, such as fast-slow and slow-slow forms, where the direction of conduction differs within the nodal pathways. The reentrant circuit is confined to the perinodal atrial tissue and to transitional cells surrounding the atrioventricular node, rather than to the compact atrioventricular node itself. Autonomic tone can significantly influence atrioventricular nodal conduction properties, and increased sympathetic activity or premature atrial beats often trigger tachycardia.[12][13]

## Histopathology

Results from histopathologic studies provide important insights into the structural basis of dual atrioventricular nodal pathway physiology observed in AVNRT. The atrioventricular node is a complex structure composed of the compact atrioventricular node, transitional cells, and surrounding atrial myocardium. Histological examinations have demonstrated heterogeneity in the arrangement of these tissues, particularly within the posterior and inferior atrial inputs to the atrioventricular node.

The so-called slow pathway is thought to correspond to inferior extensions of atrioventricular nodal tissue located near the coronary sinus ostium and the tricuspid annulus, where loosely organized transitional cells exhibit slower conduction properties. In contrast, the fast pathway is associated with superior atrial inputs near the interatrial septum, where myocardial fibers are more compact and aligned, allowing faster conduction of impulses. These anatomic differences in cellular organization, fiber orientation, and intercellular coupling create regions with distinct electrophysiologic properties, providing the substrate for dual-pathway conduction and the development of reentrant circuits characteristic of AVNRT.[14][15][16]

## History and Physical

Patients with AVNRT usually present with symptoms of rapid onset, rapid offset, and palpitations, which may be associated with dizziness, syncope, and shortness of breath. In older adults, the intermittent palpitations may accompany chest pain. Less common symptoms may include anxiety and polyuria secondary to atrial natriuretic factor, secreted mainly by the atria in response to atrial stretch.

Patients with AVNRT and a known history of coronary artery disease may present with a myocardial infarction, while those with a known history of congestive heart failure may present with an acute exacerbation and possibly reduced ejection fraction secondary to tachycardia-induced cardiomyopathy.[17] Although syncope is uncommon in patients with AVNRT, it is typical in those with a heart rate greater than 170 beats per minute (bpm), because impaired ventricular filling reduces cardiac output and cerebral perfusion. Depression of the sinoatrial node secondary to tachycardia also may contribute to syncopal symptoms in patients who present with AVNRT.[18]

Patients with AVNRT usually present with a heart rate between 140 and 280 bpm. During a physical examination, a clinician might be able to appreciate cannon A waves as the atrium contracts in conjunction with the ventricular contractions against a closed tricuspid valve. Hypotension secondary to decreased ventricular filling also may be a significant physical examination finding in patients presenting with AVNRT.[19]

## Evaluation

Evaluation of a patient with suspected AVNRT begins with a careful clinical history and physical examination. Patients typically present with sudden-onset palpitations that start and terminate abruptly, often accompanied by dizziness, dyspnea, chest discomfort, or anxiety. A 12-lead electrocardiogram obtained during tachycardia is the most important diagnostic tool and usually demonstrates a regular narrow-complex tachycardia with rates between 140 and 250 bpm, often with absent or retrograde P waves (see **Image**. Electrocardiogram of Typical AVNRT With Pseudo-S Waves). When the arrhythmia is not captured on a resting electrocardiogram, ambulatory monitoring such as Holter monitoring or event recorders may help document intermittent episodes.

An echocardiogram is not necessary, but can be considered in a select group of patients, especially when structural heart disease is suspected. Laboratory evaluation is generally directed toward identifying reversible triggers, including electrolyte abnormalities or thyroid dysfunction. In uncertain cases, an electrophysiology study may be performed to confirm dual atrioventricular nodal physiology, establish the diagnosis, and provide the opportunity for catheter ablation therapy.[20][21]

## Treatment / Management

Management of AVNRT depends on the patient's hemodynamic stability and the frequency or severity of symptoms. Treatment strategies include acute termination of the tachycardia in the emergency setting and long-term therapies aimed at preventing recurrence.[22][23] Acute management focuses on the prompt termination of the arrhythmia.

In patients who are hemodynamically stable, vagal maneuvers such as the Valsalva maneuver or carotid sinus massage are recommended as the initial intervention because they increase parasympathetic tone and transiently slow atrioventricular nodal conduction or produce transient atrioventricular nodal block. If vagal maneuvers fail, intravenous adenosine is considered the first-line pharmacologic therapy because of its rapid onset and short half-life, which transiently blocks atrioventricular nodal conduction and interrupts the reentrant circuit.[24][25] When adenosine is contraindicated or ineffective, intravenous nondihydropyridine calcium channel blockers (verapamil or diltiazem) or β-blockers may be administered to slow atrioventricular nodal conduction and terminate the tachycardia. In patients with hemodynamic instability, immediate synchronized electrical cardioversion is indicated to restore sinus rhythm.[26]

Long-term management aims to reduce recurrent episodes and improve quality of life. For patients with infrequent or minimally symptomatic episodes, conservative management or intermittent use of vagal maneuvers may be sufficient. Pharmacologic therapy with oral β-blockers, verapamil, or diltiazem can be used to prevent recurrence by suppressing atrioventricular nodal conduction.

However, medications may have limited efficacy and potential adverse effects. Consequently, catheter ablation of the slow pathway has become the preferred definitive therapy for symptomatic AVNRT.[22][23] Radiofrequency ablation targets the slow pathway, eliminating the substrate for reentry while preserving normal atrioventricular nodal conduction.

Catheter ablation has a success rate exceeding 95%, with a low risk of complications, including a small risk of complete atrioventricular block requiring pacemaker implantation. Please see StatPearls' companion reference, "Electrophysiology Study and Ablation of Atrioventricular Nodal Tachycardia," for further information. Therefore, current guidelines recommend catheter ablation as first-line therapy for many symptomatic patients, particularly those with recurrent or drug-refractory episodes.

## Differential Diagnosis

Once a narrow QRS complex tachycardia is present on an electrocardiogram, and the rhythm is determined to be regular, differential rhythms include the following:

- Atrioventricular nodal reentrant tachycardia
- Atrioventricular reentrant tachycardia
- Intra-atrial reentrant tachycardia
- Sinoatrial nodal reentrant tachycardia
- Junctional ectopic tachycardia
- Atrial tachycardia [27]

## Prognosis

The prognosis of AVNRT is generally excellent, particularly in individuals without structural heart disease. The arrhythmia is typically benign and rarely associated with increased mortality. Most patients experience recurrent but well-tolerated episodes of palpitations, and long-term outcomes are favorable with appropriate treatment.

Catheter ablation of the slow pathway has become the definitive therapy, with success rates exceeding 95%, low recurrence rates, and minimal complications. As a result, most patients achieve durable symptom relief and an excellent quality of life following treatment. In rare cases, patients may develop other atrial arrhythmias, such as atrial fibrillation, during long-term follow-up.[28]

## Complications

Although AVNRT is generally considered a benign arrhythmia, several complications may occur, particularly in patients with frequent or prolonged episodes. Persistent tachycardia can lead to hemodynamic instability, presenting with hypotension, presyncope, or syncope due to reduced ventricular filling and decreased cardiac output. Recurrent episodes may also cause tachycardia-induced cardiomyopathy, especially when the arrhythmia remains undiagnosed or inadequately treated for prolonged periods.

Patients with underlying structural heart disease may experience worsening heart failure symptoms during sustained tachycardia. Additionally, frequent symptomatic episodes can significantly impair quality of life, leading to anxiety,

reduced exercise tolerance, and repeated health care utilization. Complications may also arise from therapeutic interventions; for example, pharmacologic therapy with atrioventricular nodal blocking agents may result in bradycardia or hypotension, while catheter ablation, although highly effective, carries a small risk of atrioventricular block that may require permanent pacemaker implantation. Overall, serious complications are uncommon, and most patients have excellent outcomes with appropriate management.[22][23]

## Deterrence and Patient Education

Patient education plays an important role in the management of AVNRT. Patients should be informed that AVNRT is generally a benign arrhythmia with an excellent prognosis, particularly in the absence of structural heart disease. Education should focus on recognizing common symptoms such as sudden-onset palpitations, dizziness, or shortness of breath, and understanding when to seek medical attention. Patients can also be taught simple vagal maneuvers, such as the Valsalva maneuver, which may help terminate tachycardia episodes.

Avoidance of potential triggers, including excessive caffeine, alcohol, stimulant medications, and emotional stress, may reduce the frequency of episodes in some individuals. Patients with recurrent or symptomatic episodes should be counseled about the effectiveness and safety of catheter ablation as a definitive treatment option. Additionally, patients should be encouraged to maintain regular follow-up with their health care professional and adhere to prescribed medications when medical therapy is used.

## Pearls and Other Issues

In rare instances, atrioventricular nodal reentry tachycardia can result in sudden cardiac arrest.[29]

## Enhancing Healthcare Team Outcomes

Optimal management of AVNRT relies on coordinated interdisciplinary teamwork to ensure rapid diagnosis and effective treatment. Patients typically present with sudden-onset palpitations and a regular narrow-complex tachycardia, and timely recognition is essential to initiate appropriate therapy. Patients with hemodynamic instability require immediate synchronized cardioversion, while patients who are stable are generally treated with a stepwise approach beginning with vagal maneuvers followed by pharmacologic therapy such as adenosine or other atrioventricular nodal blocking agents.

Nursing staff play a key role in obtaining rapid electrocardiograms, initiating vagal maneuvers, establishing intravenous access, and monitoring for complications. Pharmacists help verify contraindications to medications and ensure appropriate dosing and administration of antiarrhythmic medications. Clinicians are responsible for accurate rhythm diagnosis, assessment of hemodynamic stability, and treatment planning. Early involvement of electrophysiology specialists facilitates timely evaluation for catheter ablation, a highly effective, definitive therapy that improves symptom control and overall patient outcomes.

## Review Questions

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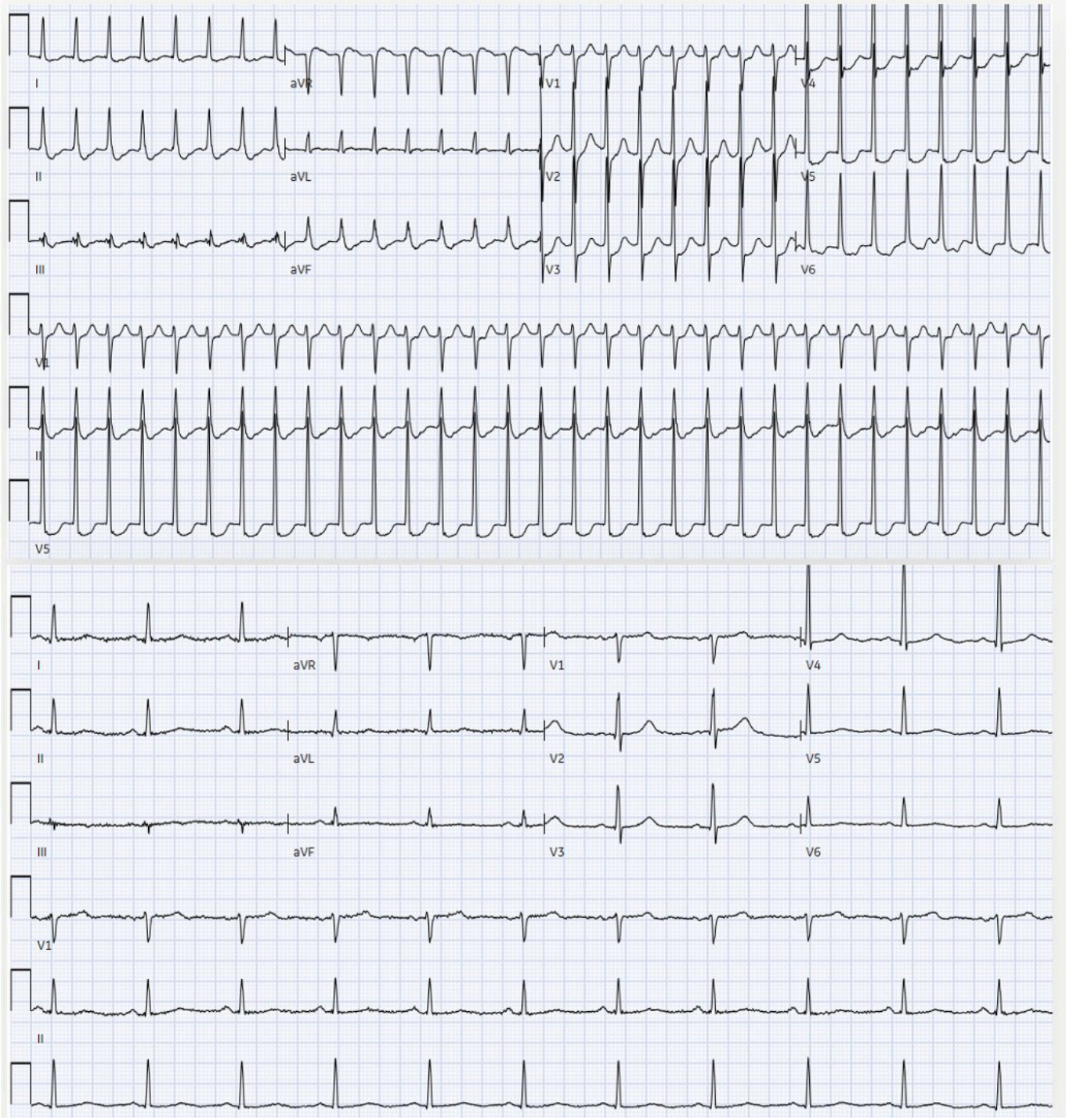
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## Figures




### Figure 2

Electrocardiogram of Typical AVNRT With Pseudo-S Waves. This 12-lead electrocardiogram demonstrates typical (slow-fast) atrioventricular nodal reentrant tachycardia (top panel) featuring characteristic terminal pseudo-S waves, representing retrograde P-wave activity, which are absent in the baseline sinus rhythm (bottom panel). Contributed by I Ahmed, MD

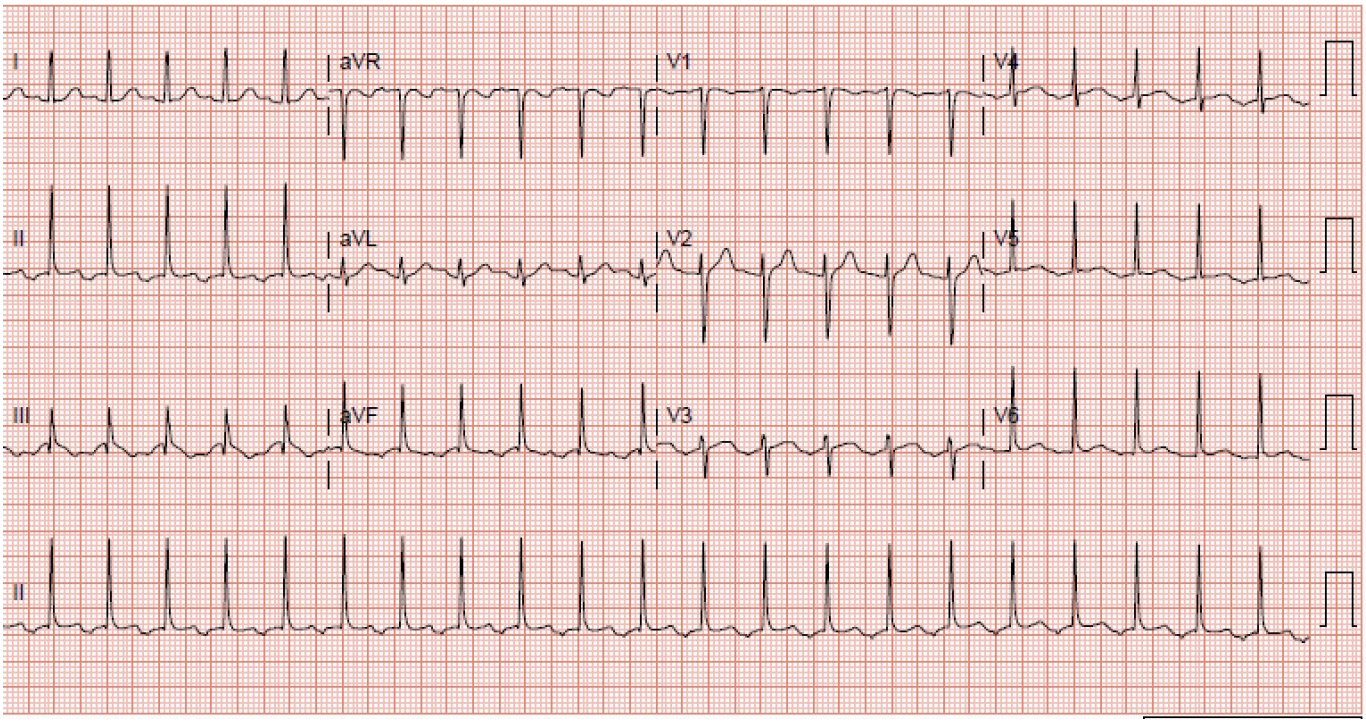



**Figure 1**

Electrocardiogram of a Long RP Narrow Complex Tachycardia. This 12-lead electrocardiogram demonstrates a regular, narrow-complex supraventricular tachycardia with a long RP interval and retrograde P-waves (inverted in leads II, III, and aVF; upright in V1). Contributed by Y Hafeez, DO

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